Leibniz's Notation

Solve for $\frac{d}{dx}(x^2)$

$$\begin{aligned} d(x^2) &= \lim_{h\to 0}[(x+h)^2 - x^2] \\ &= (x+dx)^2 - x^2 \\ &= \cancel{x^2 - x^2} + 2xdx + \cancel{dx^2} \\ &= 2xdx \\ \frac{d}{dx}[x^2] &= 2x \end{aligned}$$

Prove the product rule

$$\begin{aligned} d[f(x)*g(x)] &= \lim_{h\to 0}[f(x+h)*g(x+h) - f(x)g(x)] \\ &= (f(x)+df)(g(x)+dg) - f(x)g(x) \\ &= \cancel{f(x)g(x) - f(x)g(x)} + f(x)dg + g(x)df + \cancel{dfdg} \\ &= f(x)dg + g(x)df \\ \frac{d}{dx}[f(x)*g(x)] &= f(x)\frac{dg}{dx} + g(x)\frac{df}{dx} \end{aligned}$$

A Mope Explicit Notation

$$\delta_{\alpha\ldots}(\beta\ldots) = \delta\{\alpha\ldots\}(\beta\ldots) = \text{Leibniz } d(\alpha\ldots)$$

$$D\{\alpha\ldots\}(\beta\ldots) = \frac{\delta\{\alpha\ldots\}(\beta\ldots)}{\delta_{\beta\ldots}} = \text{Leibniz } \frac{d}{d\beta\ldots}(\alpha\ldots)$$

Solve for $D\{x^2\}(x)$

$$\begin{aligned} f &= x^2 \\ \delta_f(x) &= \lim_{h\to 0}(f(x+h) - f(x)) \\ &= \lim_{\delta_x\to 0}\left((x+\delta_x)^2 - x^2\right) \\ &= (x+\delta_x)^2 - x^2 \\ &= \cancel{x^2 - x^2} + 2x\delta_x + \cancel{\delta_x^2} \\ &= 2x\delta_x \\ \frac{\delta\{x^2\}(x)}{\delta_x} &= 2x \\ D\{x^2\}(x) &= 2x \end{aligned}$$

Prove the product rule

$$\delta\{f(x)g(x)\}(x) = \lim_{\delta_x \to 0}[f(x+\delta_x) * g(x+\delta_x) - f(x)g(x)]$$

$$f(x+\delta_x) = f(x) + \delta_f(x)$$

$$g(x+\delta_x) = g(x) + \delta_g(x)$$

$$\begin{aligned}\delta\{f(x)g(x)\}(x) &= \big(f(x)+\delta_f(x)\big)\big(g(x)+\delta_g(x)\big) - f(x)g(x) \\ &= \cancel{f(x)g(x) - f(x)g(x)} + f(x)\delta_g(x) + g(x)\delta_f(x) + \cancel{\delta_f(x)\delta_g(x)} \\ &= f(x)\delta_g(x) + g(x)\delta_f(x)\end{aligned}$$

$$D\{f(x)g(x)\}(x) = f(x)\frac{\delta_g(x)}{\delta_x} + g(x)\frac{\delta_f(x)}{\delta_x}$$